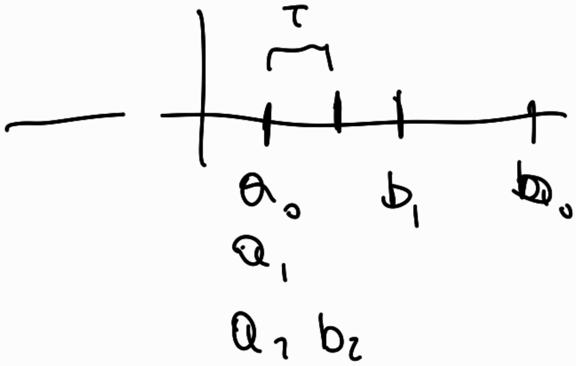
Stopping criteria

Bisection method
a priori error estimate



Stopping Newton iterations

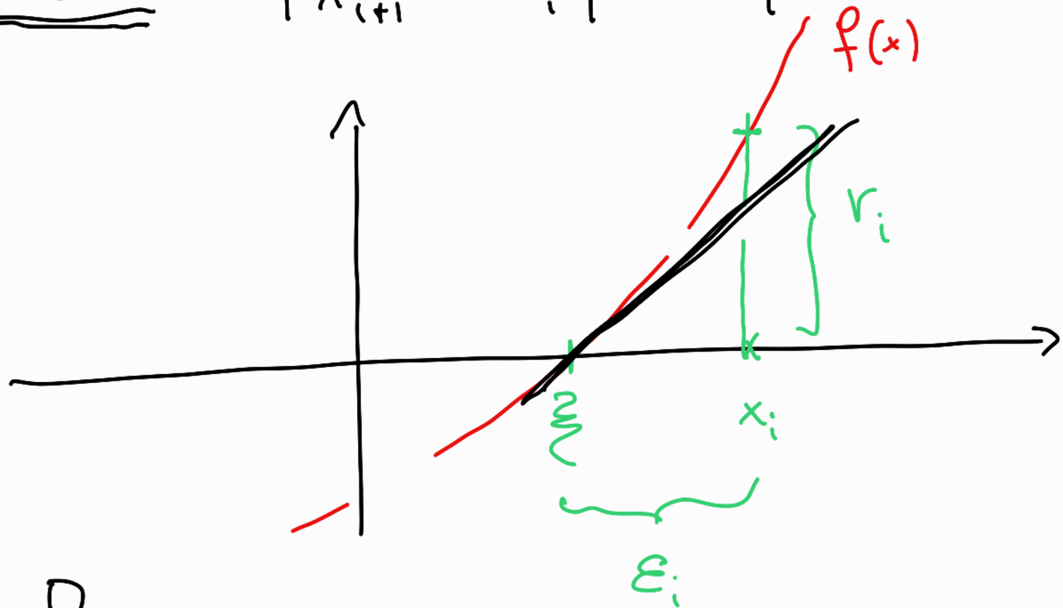
$f(\xi) = 0$

$$\varepsilon_N = |x_N - \xi| \leq \tau$$

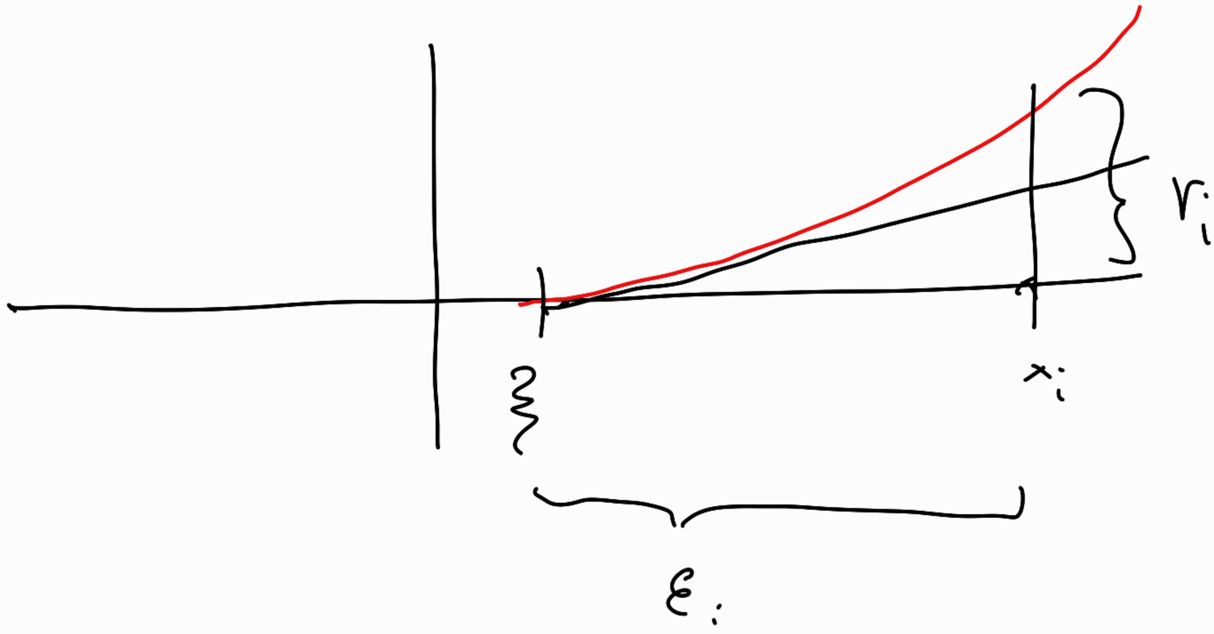
error (ε_N) — tolerance (τ)

residual $|f(x_i)| = r_i$

increment $|x_{i+1} - x_i| = \delta x_i$



$$f'(\xi) = \frac{f(x_i) - 0}{x_i - \xi} \qquad \tau > |f(x_i)| \gg |x_i - \xi| \quad \text{if } f'(\xi) \gg 0$$



if $|f'(\xi)| \ll 1$ then r_i can be much smaller than the error

For fixed point methods

find ξ st. $\underline{g(\xi) = \xi}$

$$\underline{x_{k+1} = g(x_k)}$$

$$\varepsilon_{k+1} = \xi - x_{k+1} = g(\xi) - g(x_k) =$$
$$= g'(\alpha_k)\,(\xi - x_k)$$

$\xi < \alpha_k < x_k$ or $x_k < \alpha_k < \xi$

$$\xi - x_k = \frac{1}{1 - g'(\alpha_k)} \cdot (x_{k+1} - x_k)$$

$$\text{if } g'(\alpha_k) \xrightarrow[k \to \infty]{} 0 \Leftarrow \underline{\underline{g'(\xi) = 0}}$$

$$\varepsilon_k \simeq \delta x_k$$

So for any second order method (Newton's method applied to simple roots) the increment is a good "a posteriori" estimate for the error.

Summary of non-linear equations

- Bisection method
  - Globally convergence
  - A-priori estimate of the error
  - Order 1
  - Applie only to 1D functions
- Fixed point methods
  - reformulate the root finding problem as a f.p. problem
  - Needs consistent
  - Local convergence
  - Allows constructing "fast" converging methods

    $\rightarrow$ order > 1

- Newton-Raphson is order 2 if $f'(\xi) \neq 0$

- A posteriori estimates for the error

- Stopping criteria

find ξ in $[a, b]$ such that $f(\xi) = 0$

root finding problem

find $\xi \mid g(\xi) = \xi$ fixed point problem

consistent $f(\xi) = 0 \Longleftrightarrow g(\xi) = \xi$

Fixed point iterations x_0

$x_{k+1} = g(x_k)$

convergent $x_k \xrightarrow[k \to \infty]{} \xi$

$$\lim_{k \to \infty} \frac{|x_{k+1} - \xi|}{|x_k - \xi|^p} = c \qquad p > 1$$

"high" order

Newton's method

$$g_N(x) = x - \frac{f(x)}{f'(x)}$$

A F.P. method is order 2 if

$$g'(\xi) = 0 \quad \text{and} \quad g''(\xi) \neq 0$$

For Newton's method

$$\boxed{g_N'(\xi) = 0} \iff \boxed{f'(\xi) \neq 0}$$

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ \vdots & & & & \\ a_{n1} & a_{n2} & \cdots & & a_{nn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_{b}$$

if A is diagonal : $a_{ij} = 0$ if $i \neq j$

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 & \cdots \\ 0 & a_{22} & 0 & 0 & \cdots \\ \vdots & \ddots & & & \\ & & & & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$\#$ flops $= n$

$a_{11} x_1 = b_1 \qquad x_1 = b_1 / a_{11}$

$a_{22} x_2 = b_2 \qquad x_2 = b_2 / a_{22}$

$\cdots$

if A is lower triangular

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 & 0 & \cdots \\ a_{31} & a_{32} & a_{33} \\ \vdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$a_{11} x_1 = b_1 \Rightarrow x_1 = b_1 / a_{11}$

$a_{21} x_1 + a_{22} x_2 = b_2 \Rightarrow x_2 = \frac{1}{a_{22}} (b_2 - a_{21} x_1)$

$a_{31} x_1 + a_{32} x_2 + a_{33} x_3 = b_3$

$\Leftrightarrow x_3 = \frac{1}{a_{33}} (b_3 - a_{31} x_1 - a_{32} x_2)$

$- - -$

$$x_k = \frac{1}{a_{kk}} \left(b_k - \sum_{j=1}^{k-1} a_{kj} x_j \right)$$

$$\# \text{flops} = 1 + 2 + 3 + \cdots + (n-1) + n$$

$n+1$

$n+1$

$$= (n+1) \frac{n}{2} = \frac{n^2}{2} + \frac{n}{2} \sim \frac{n^2}{2}$$